Trends in Arctic Shipping

Dr. Jackie Dawson, Full Professor & Canada Research Chair

Department of Geography, Environment and Geomatics, University of Ottawa

ESPG Environment, Society and Policy Group

uOttawa

Polar Maritime Seminar
23-24 January 2025
IMO Headquarters, London, UK

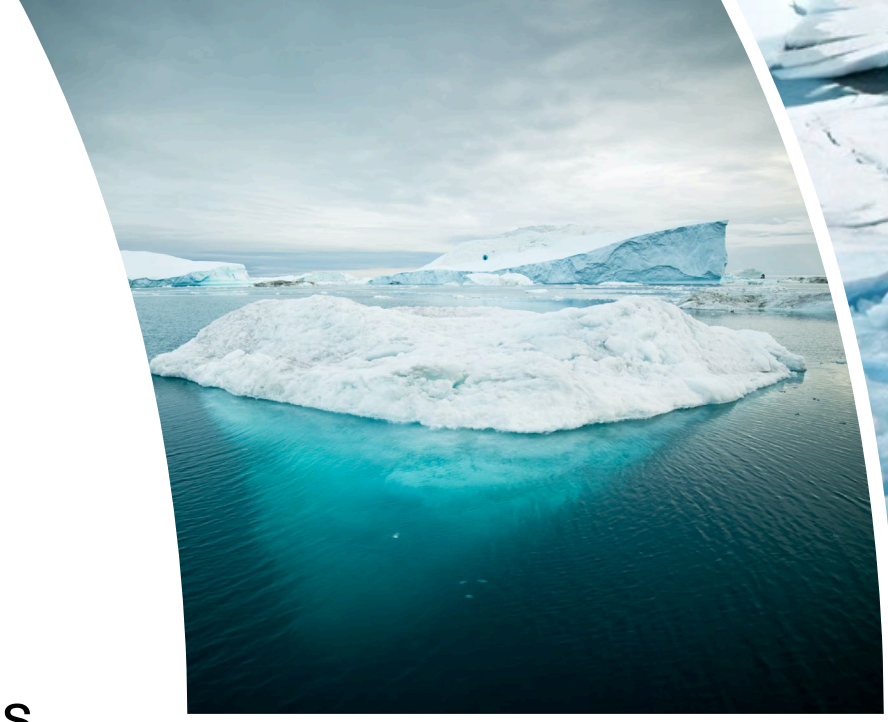
IMO INTERNATIONAL MARITIME ORGANIZATION

ARCTIC COUNCIL

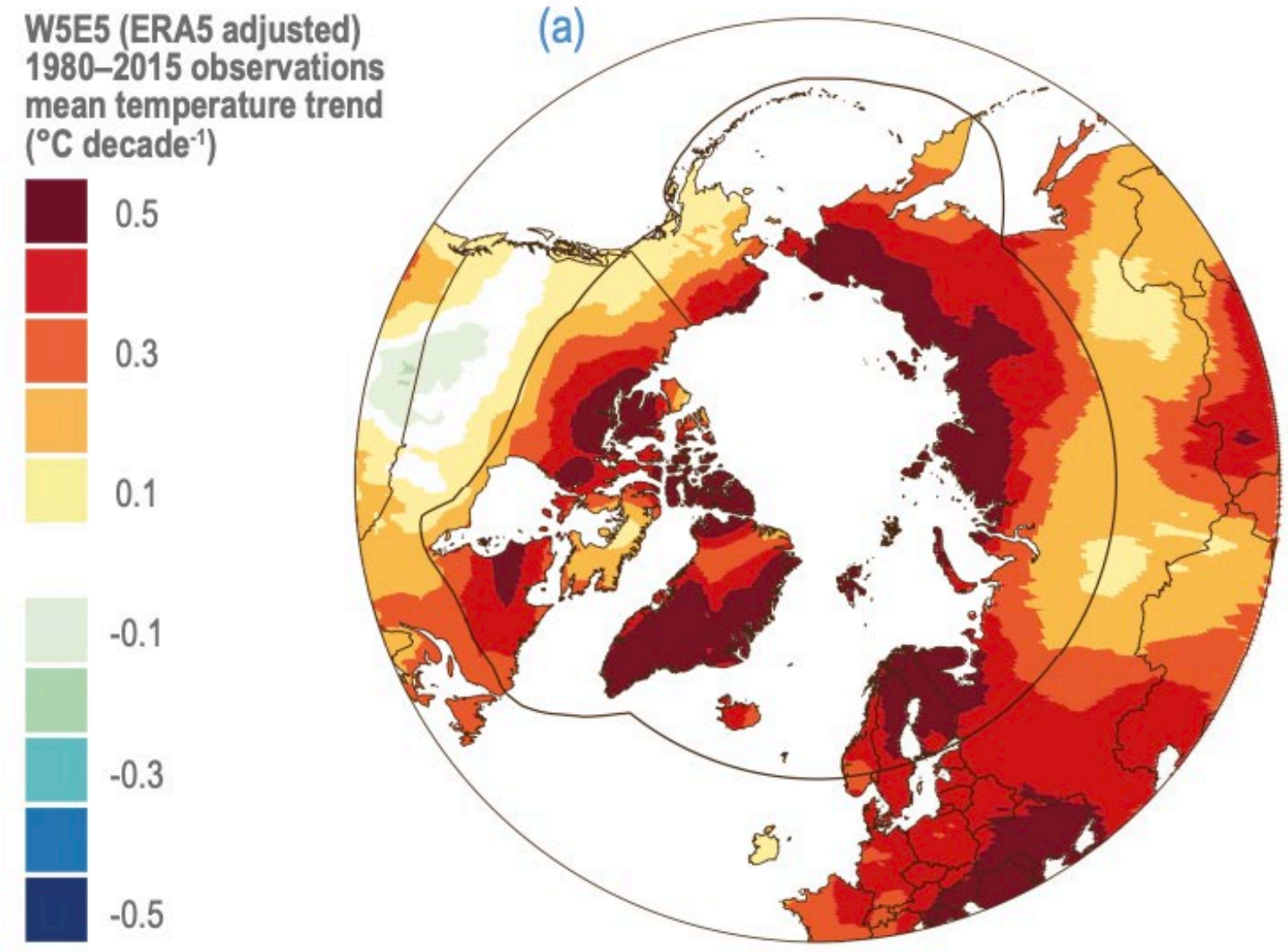
PAME Protection of the Arctic Marine Environment

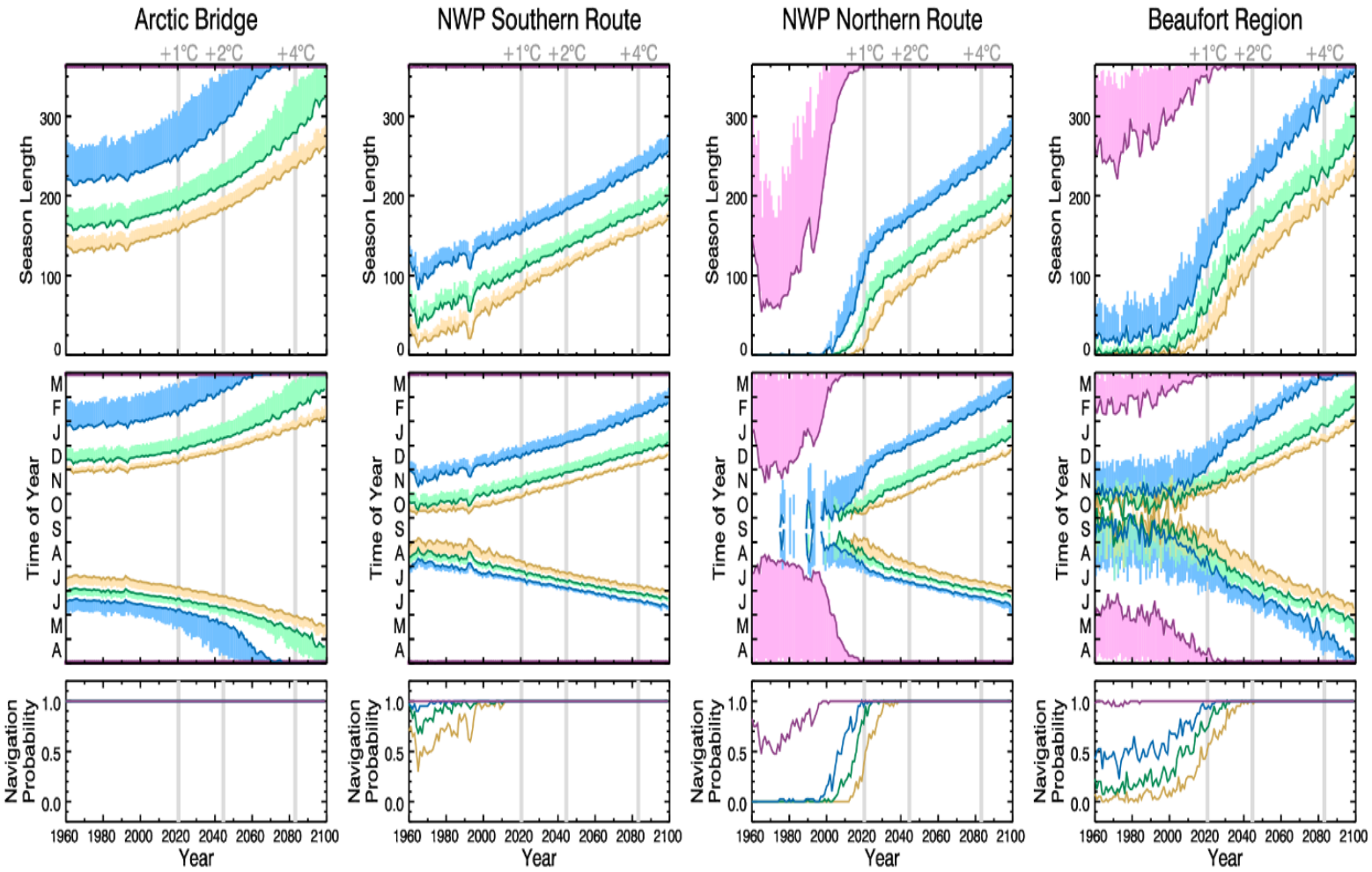
Outline

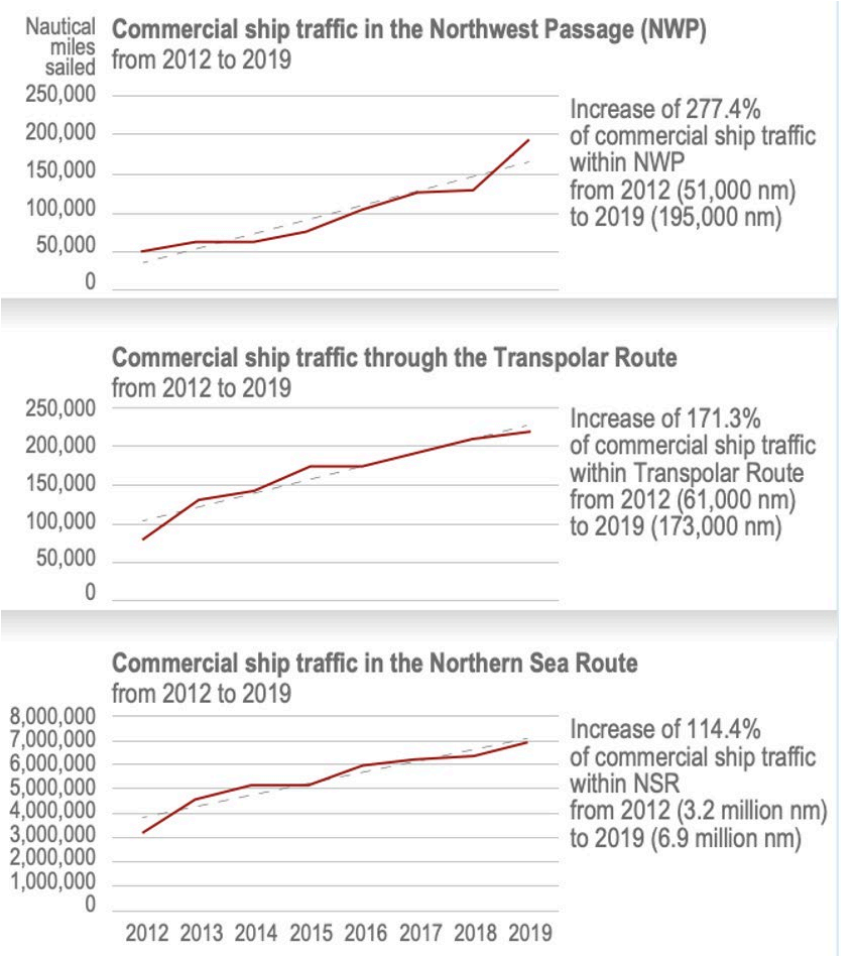
- Climate Change and Future Projections
- Global Arctic Shipping Patterns and Trends
- Ship-Ice Interactions, Accidents and Risk
- Data Challenges



Projected Warming in the Arctic

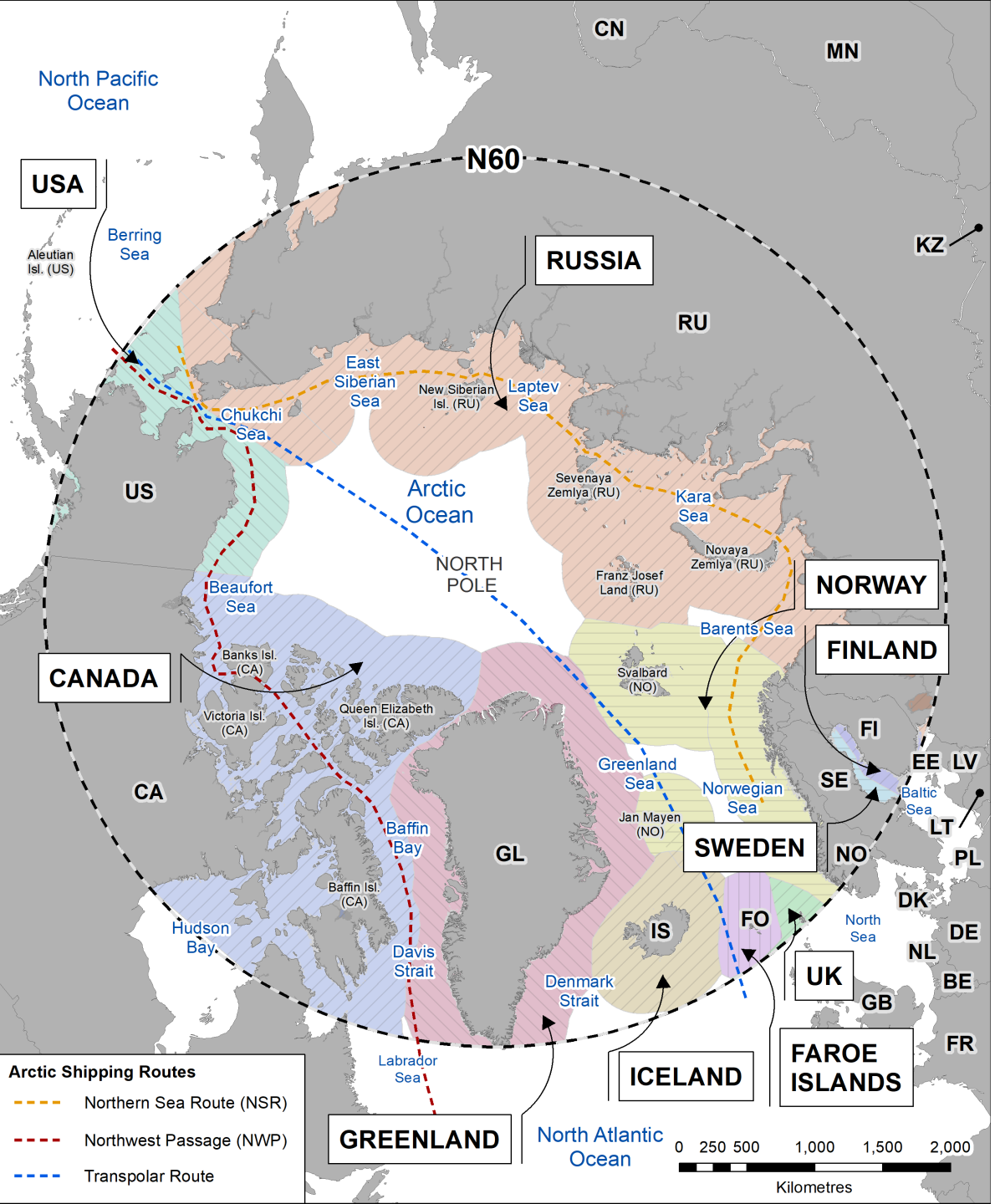




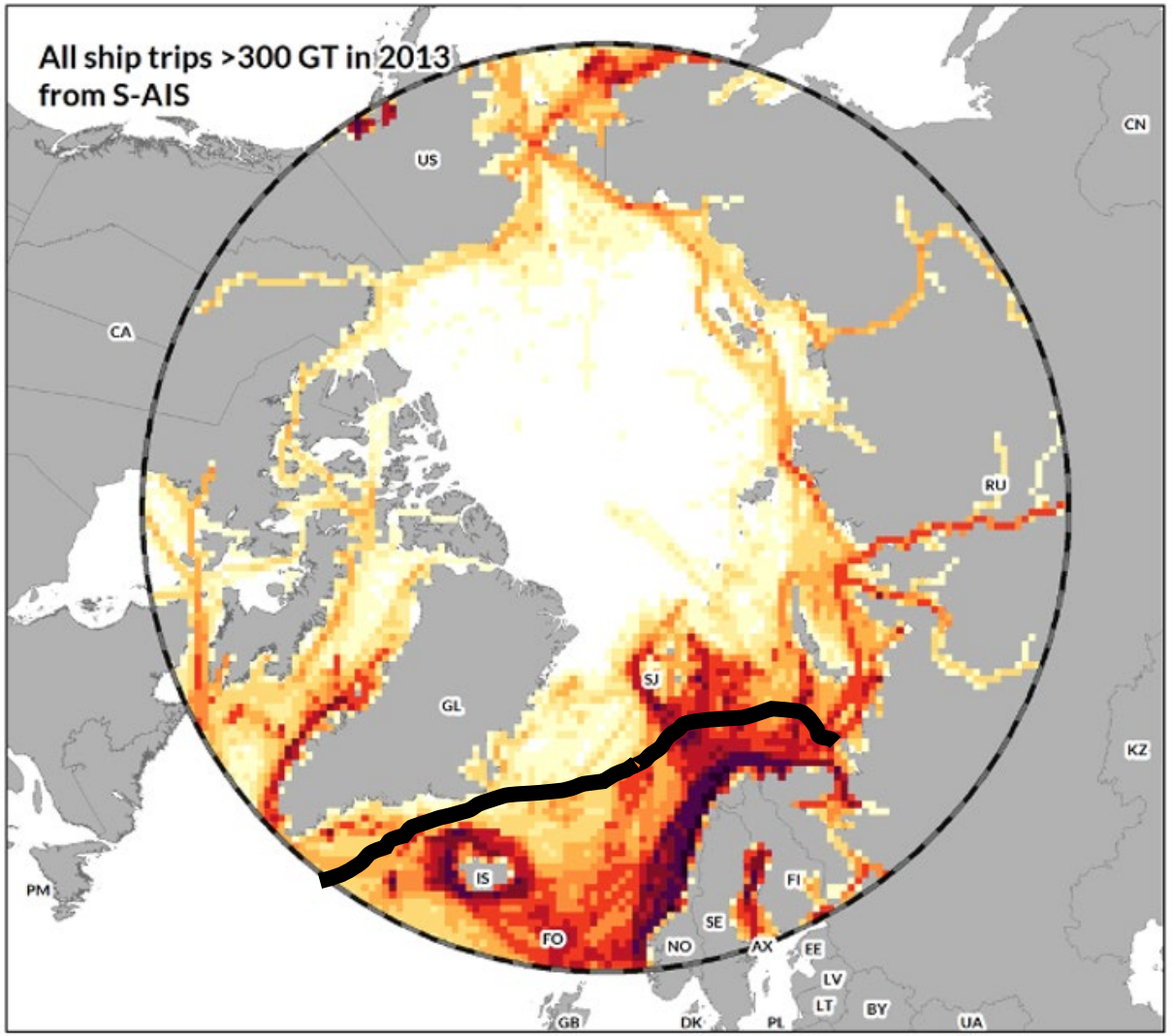


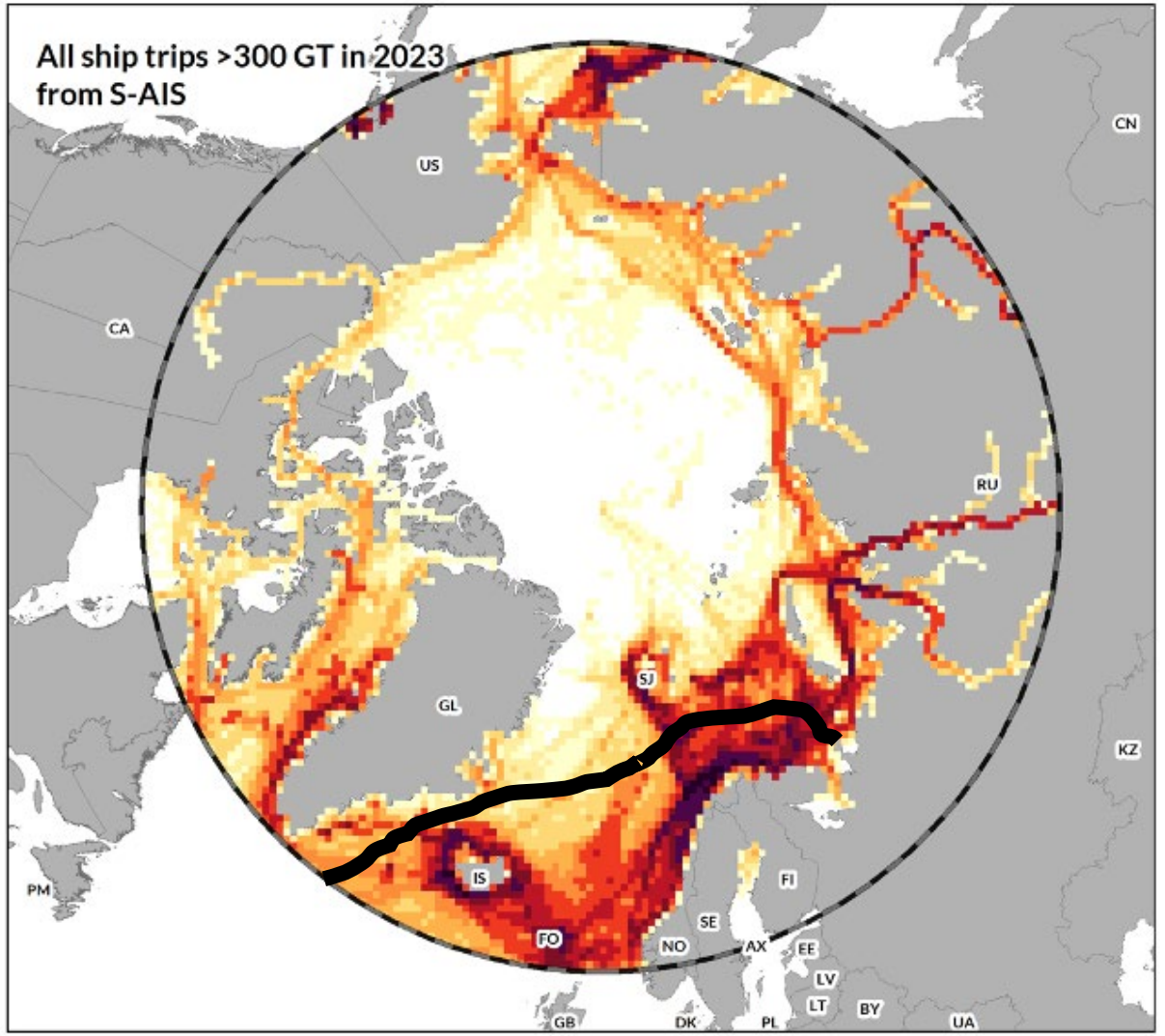
Northwest Passage: + 14 to 31 days before 4 ^{o} C warming
Northern Sea Route: 101 to 118 days annually by 2050
Transpolar Route: +56% increase in accessibility by =2050

Area of analysis (North of the 60th parallel), with Exclusive Economic Zones, North of the 60th parallel Using S-AIS (Spire/Kpler)

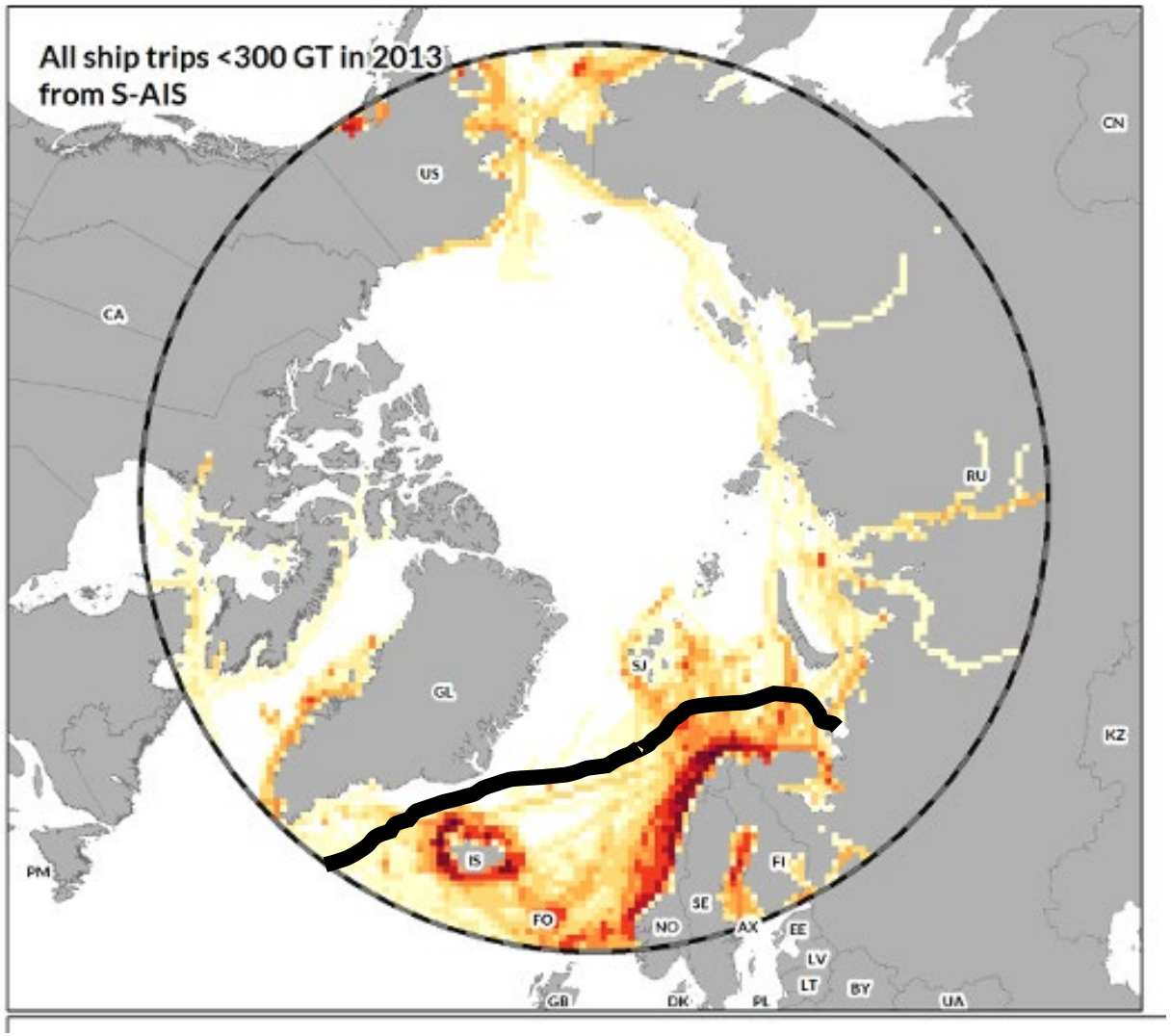


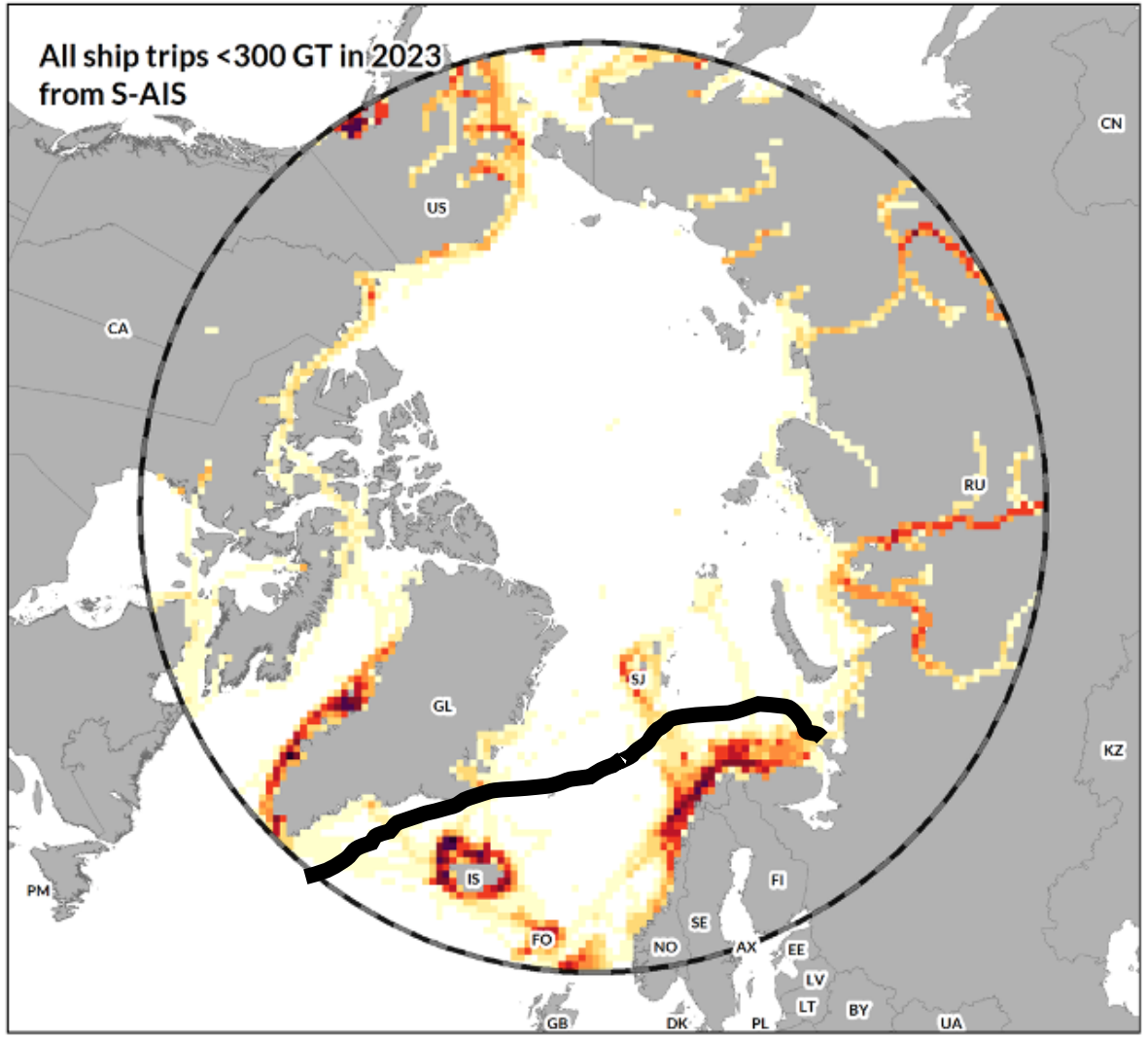
Ships Over 300 GT (2013 vs. 2023)



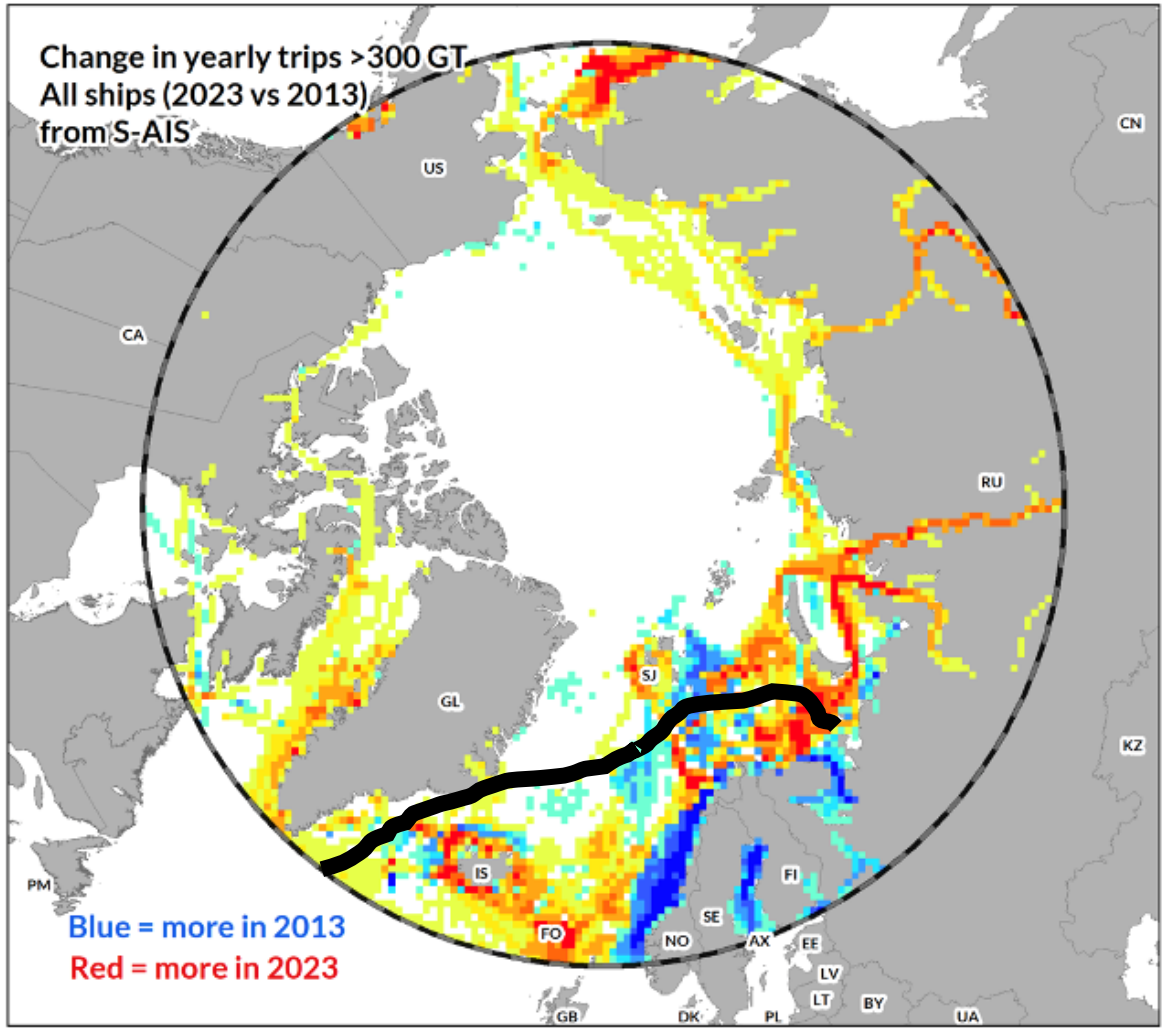


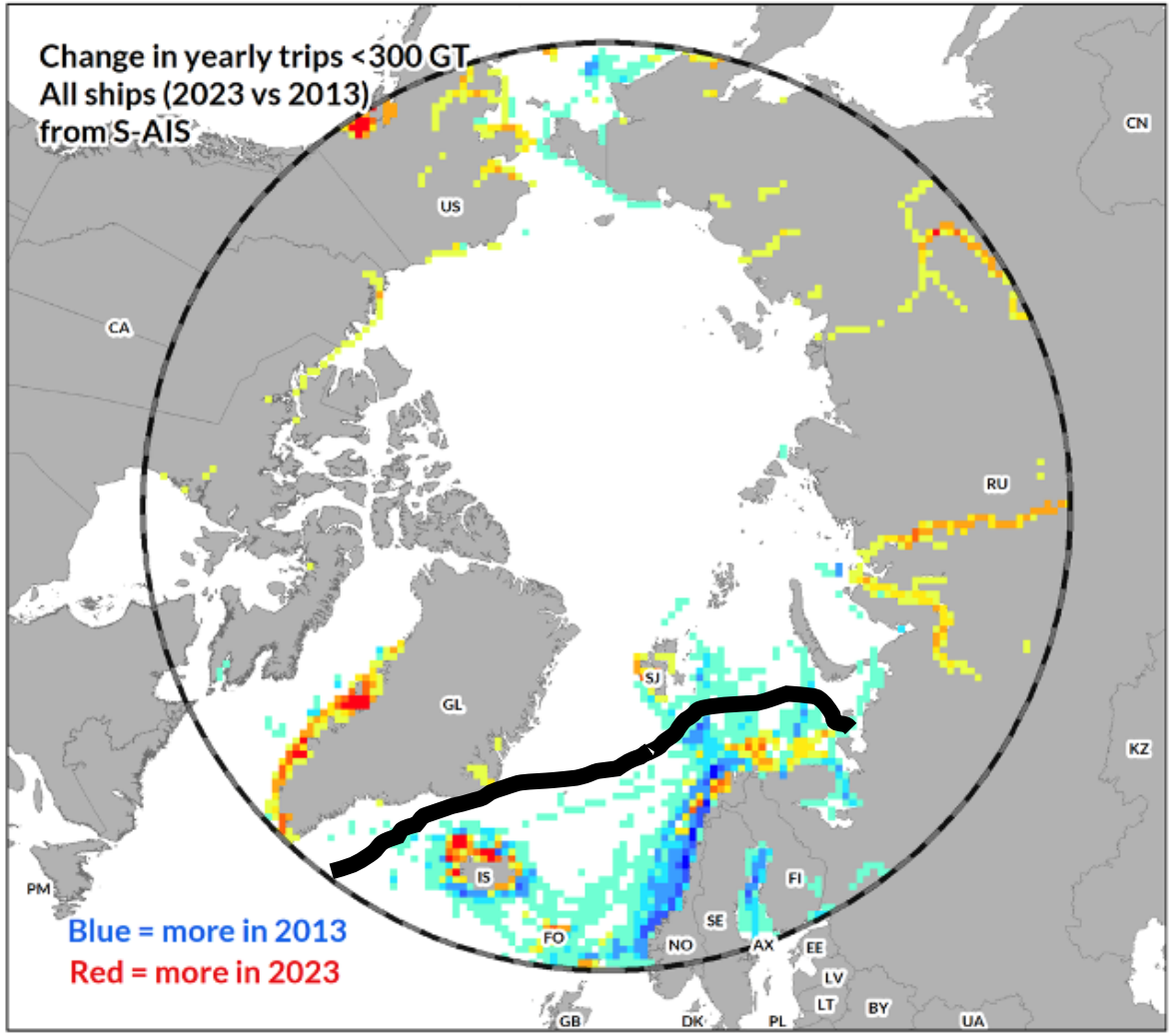
Ships Under 300 GT (2013 vs. 2023)



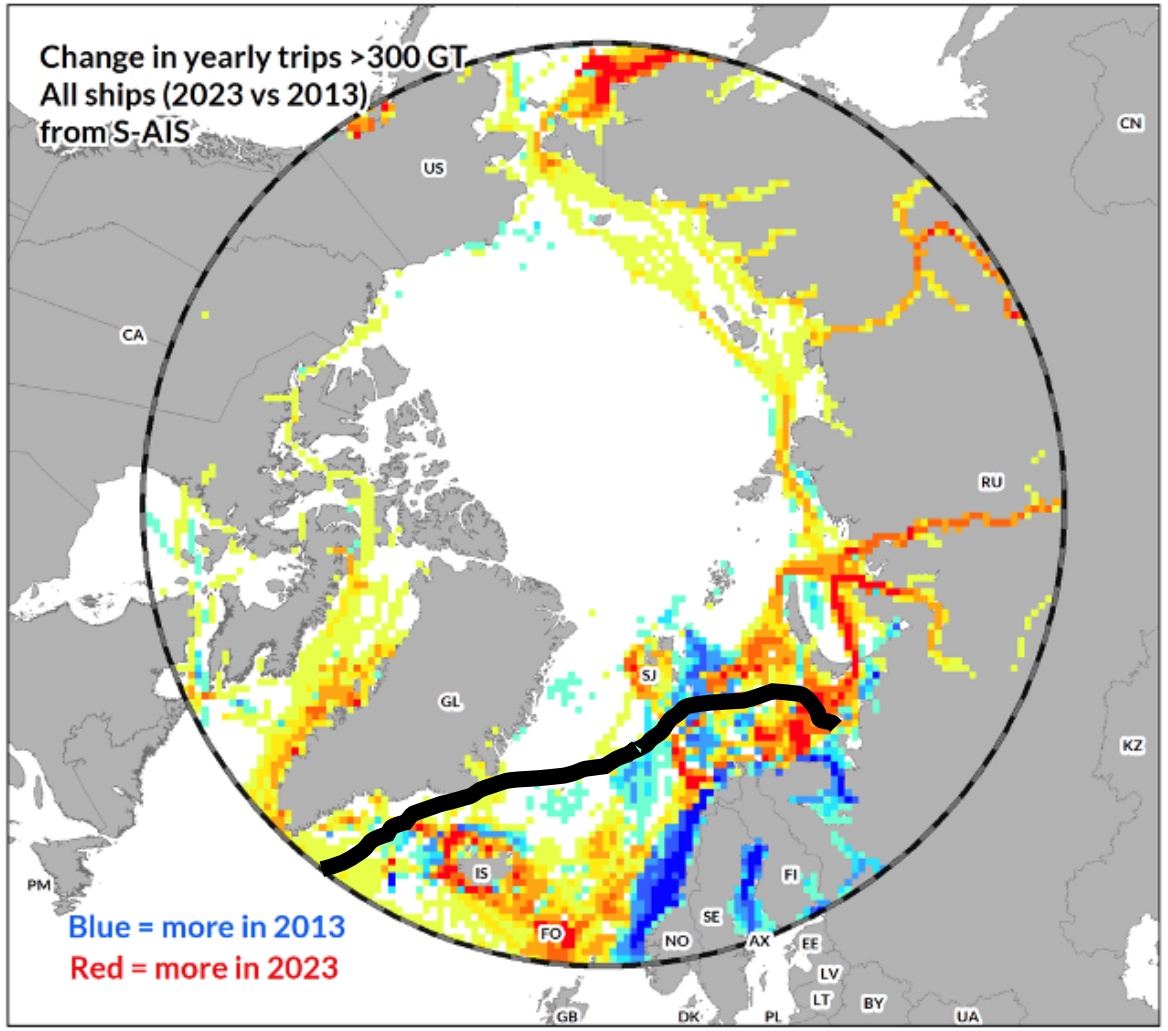


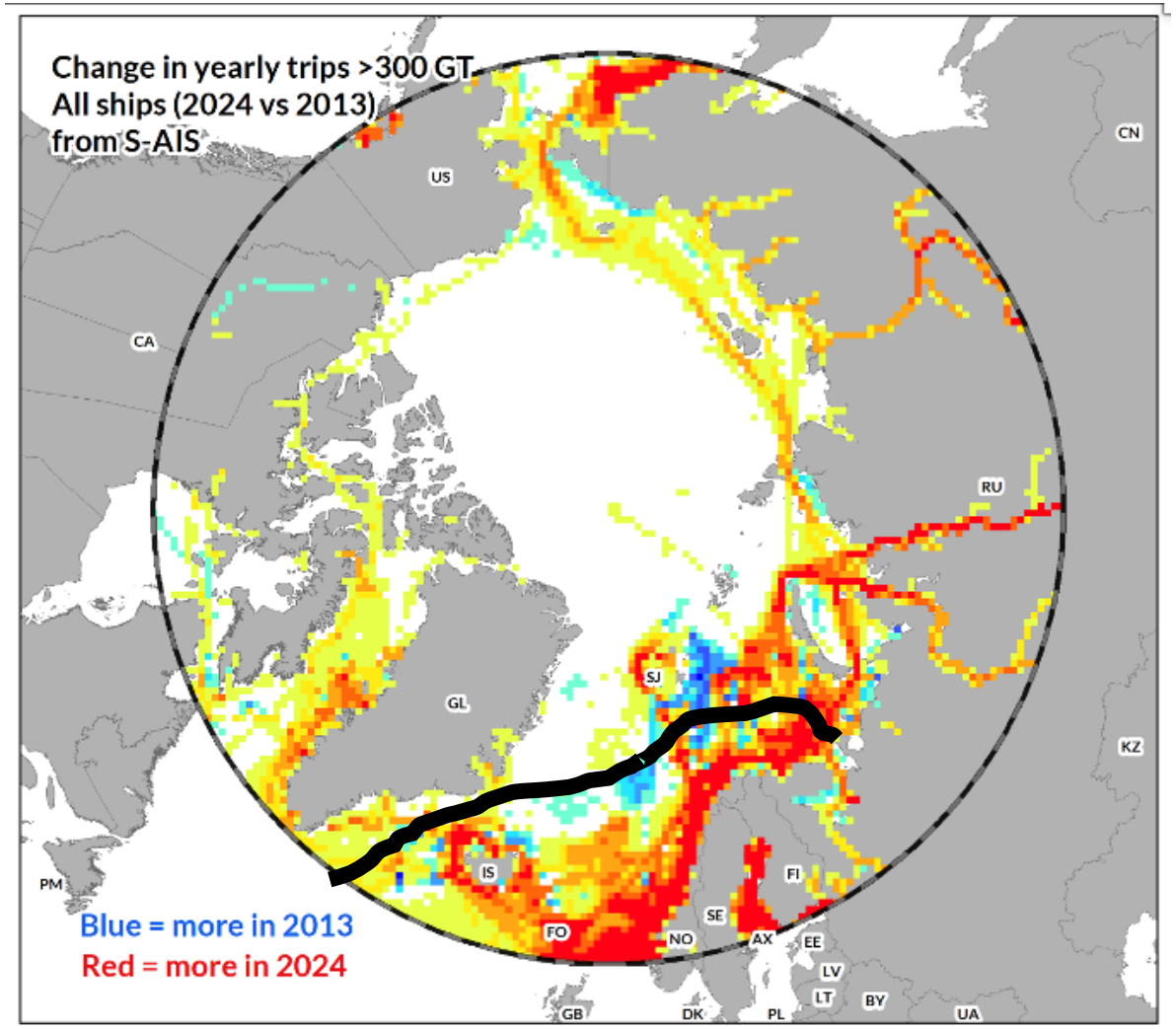
Changes in annual trips 2013 vs. 2023 (>300GT left <300GT right)



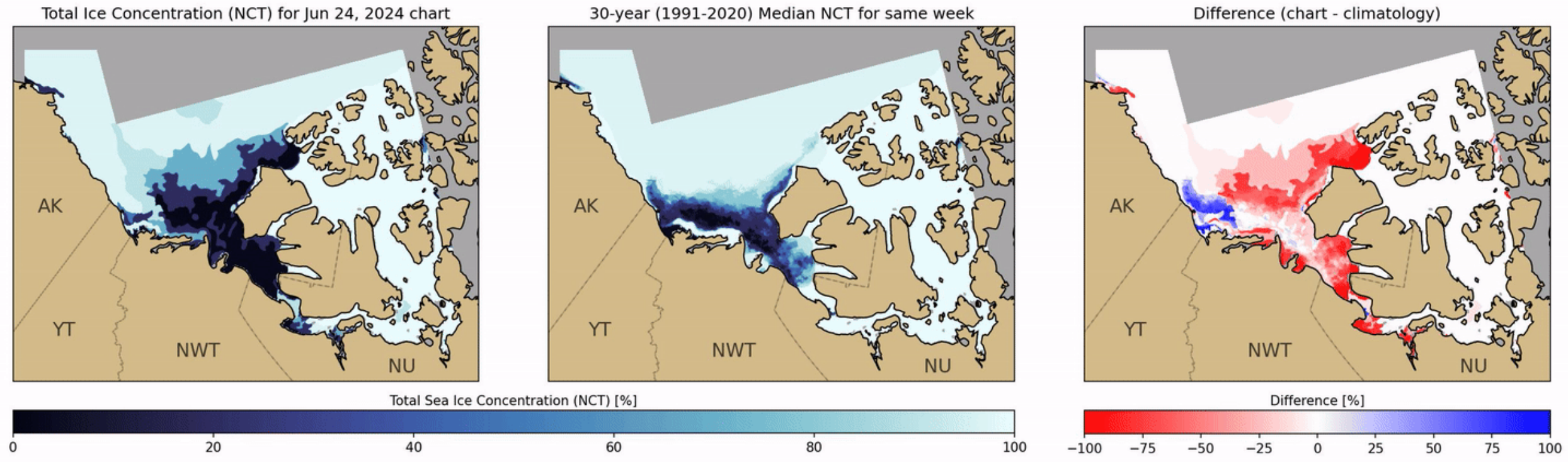


Changes in annual trips >300GT 2013 vs. 2023 (left) and 2013 vs. 2024 (right)





Animation of record low (2024) sea ice conditions in the Northen Route of the Northwest Passage

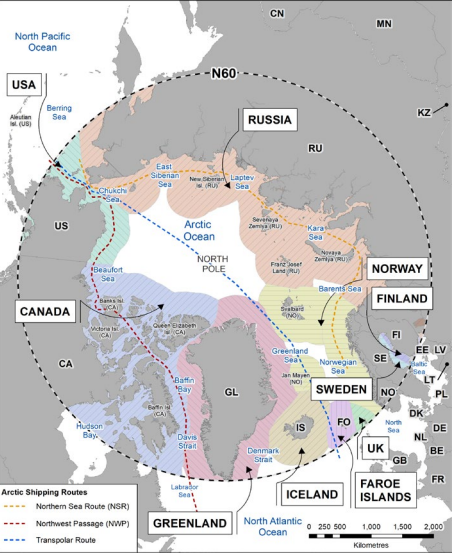


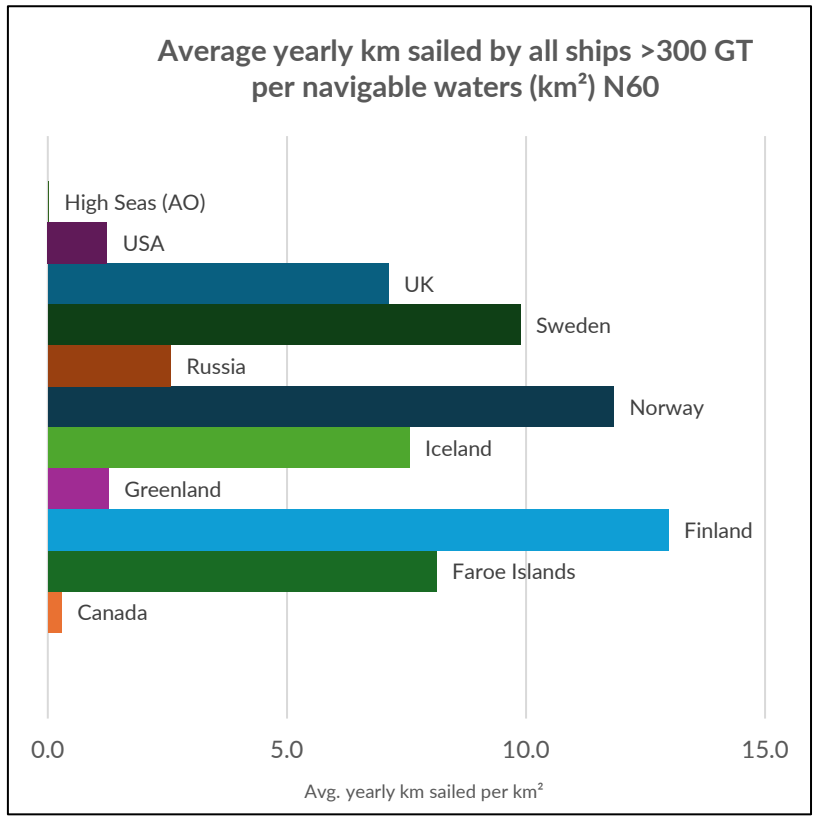
Environment and Climate Change Canada
Environnement et Changement climatique Canada

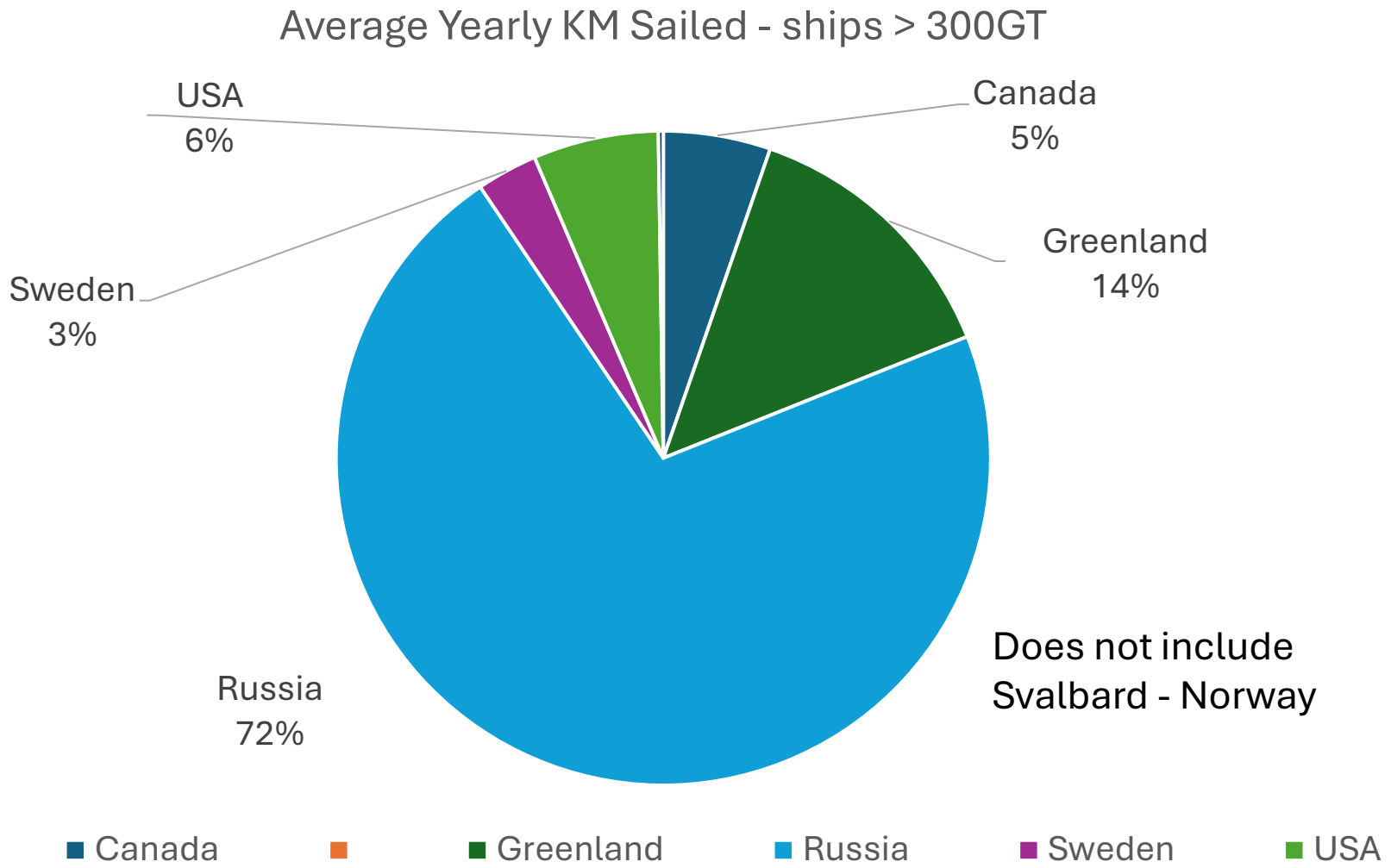
Acknowledgement - Dr. Stephen Howell

Canada

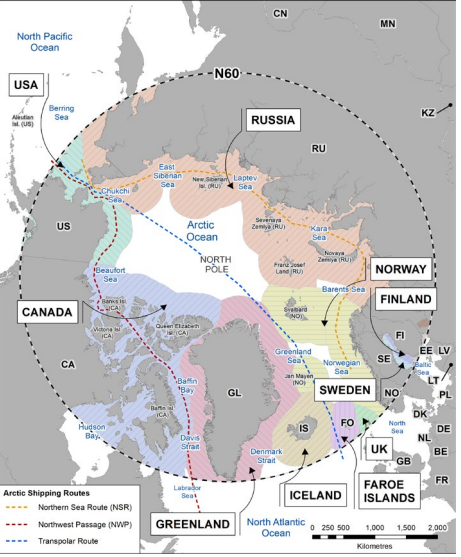
Average Annual KM Sailed (2013-2024) in EEZ of Countries in the Polar Code Area - ships > 300GT



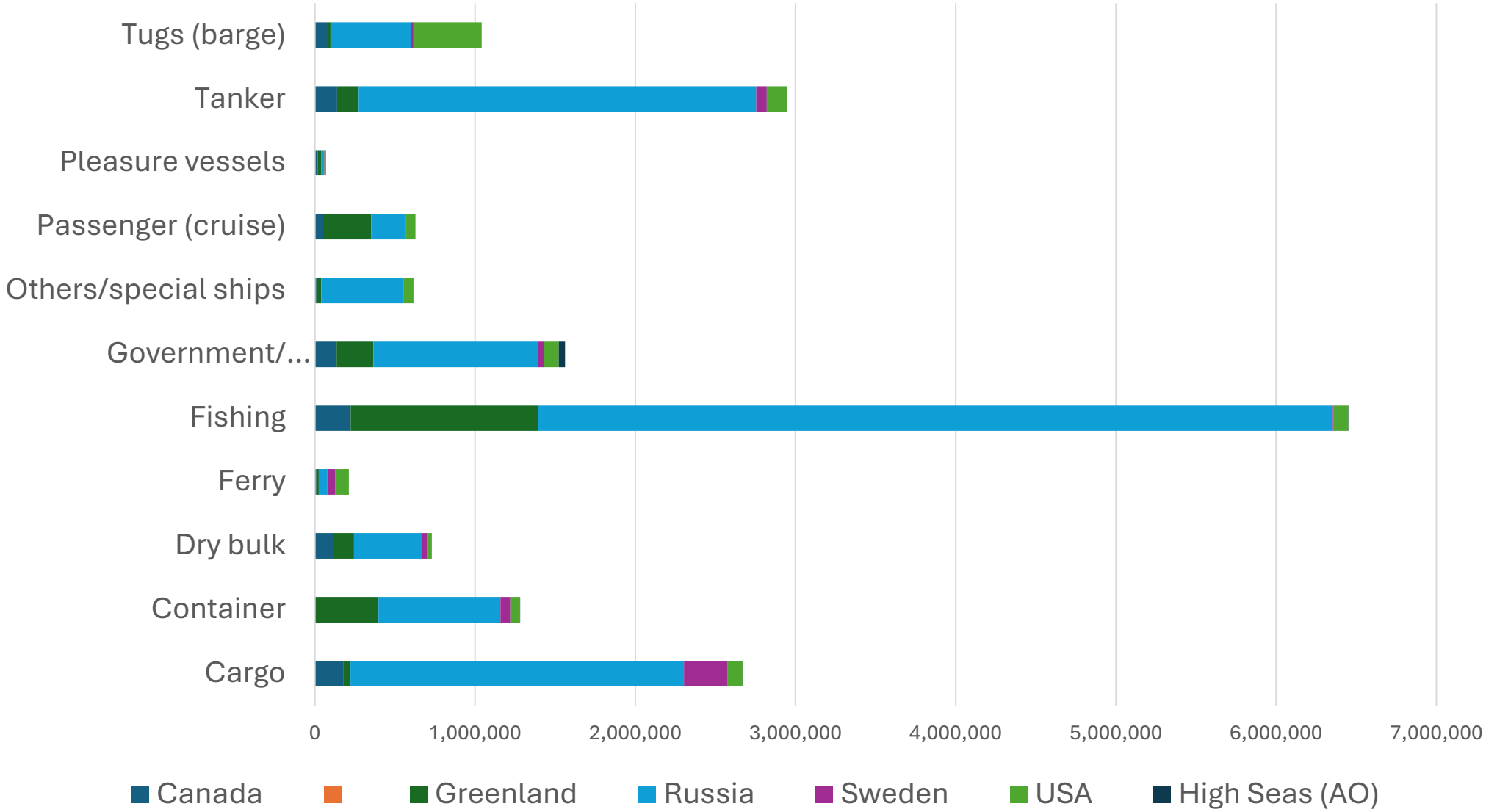


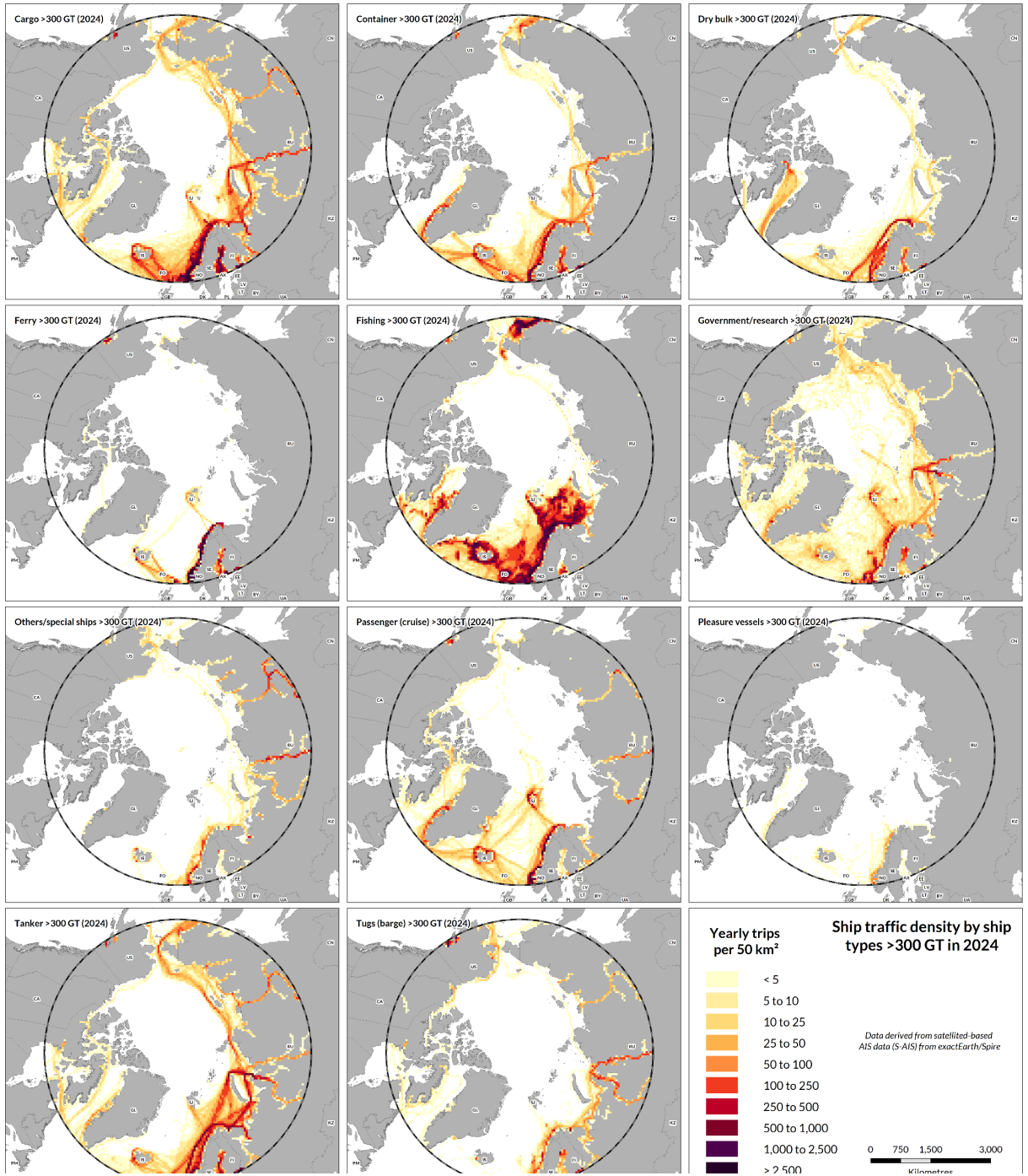


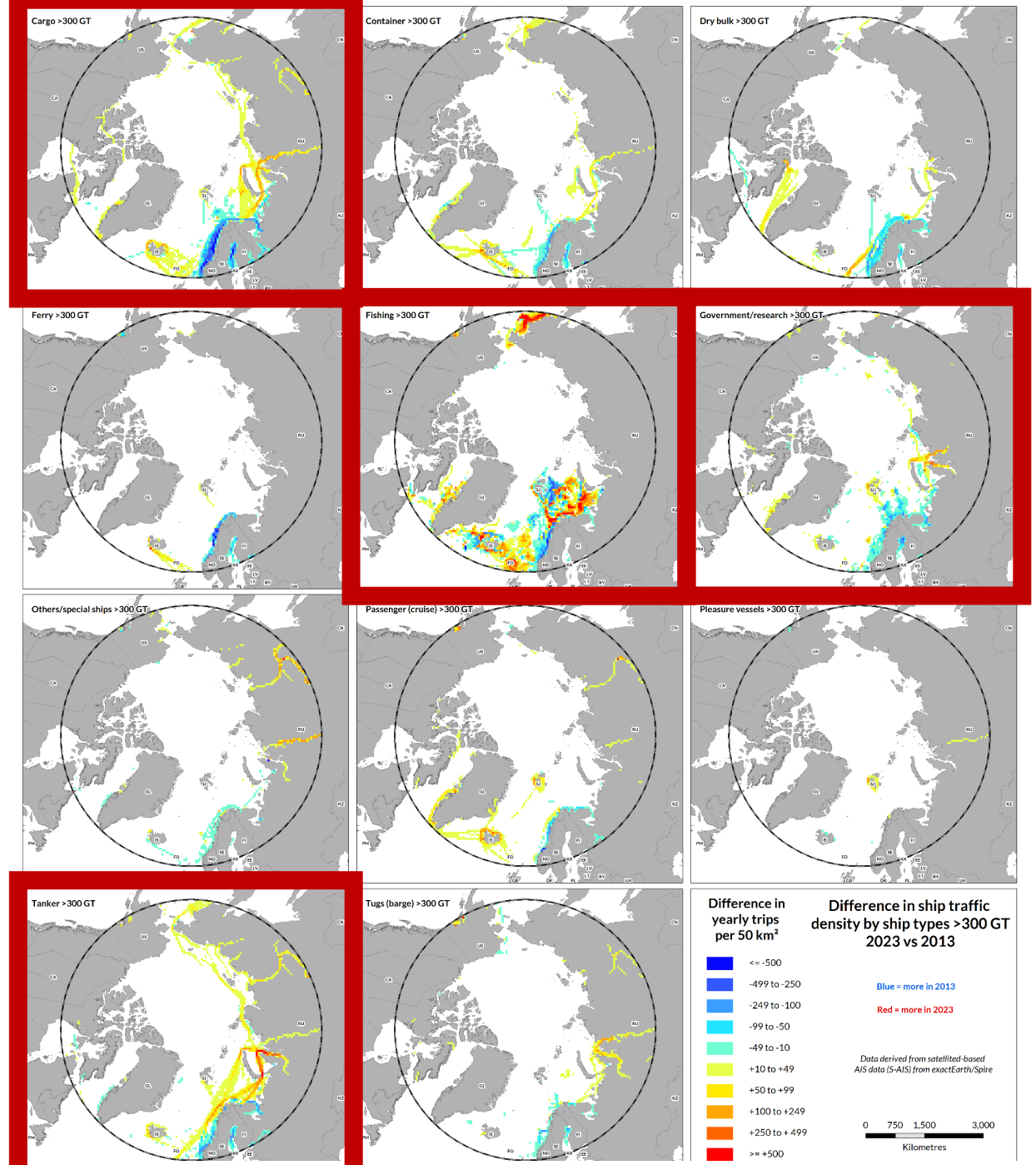
Average Annual KM Sailed (2013-2024) in EEZ of Countries in the Polar Code Area - ships > 300GT – by Vessel Type



Does not include Svalbard - Norway

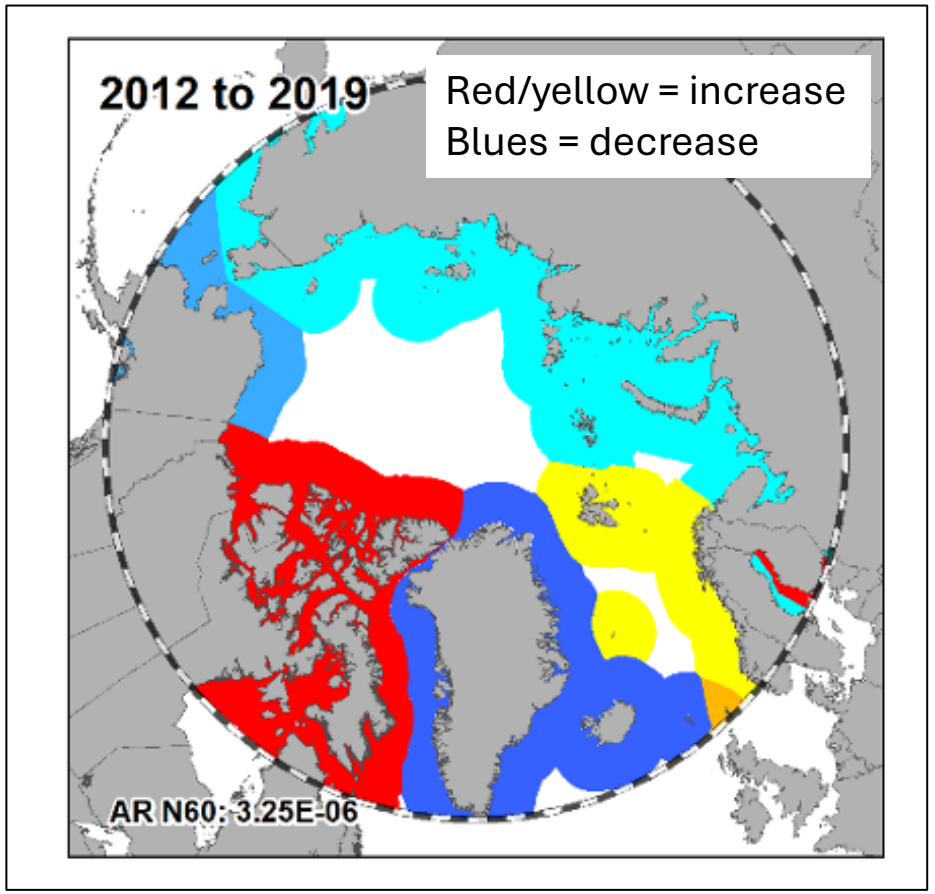


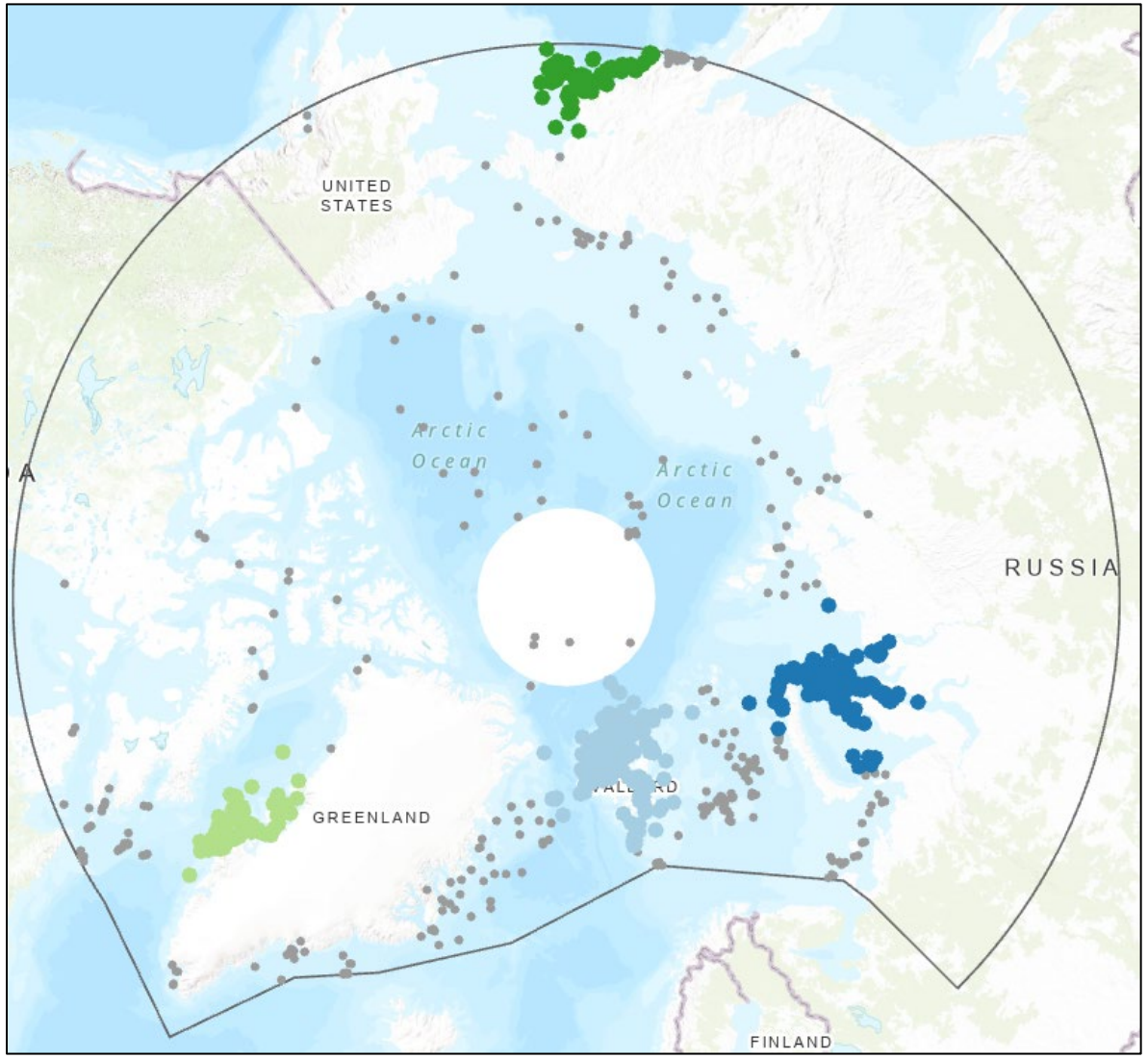

Cargo >300 GT (2024)
Container >300 GT (2024)
Dry bulk >300 GT (2024)
Ferry >300 GT (2024)
Fishing >300 GT (2024)
Government/research >300 GT (2024)
Others/special ships >300 GT (2024)
Passenger (cruise) >300 GT (2024)
Pleasure vessels >300 GT (2024)
Tanker >300 GT (2024)
Tugs (barge) >300 GT (2024)
Yearly trips per 50 km²
< 5
5 to 10
10 to 25
25 to 50
50 to 100
100 to 250
250 to 500
500 to 1,000
1,000 to 2,500
> 2,500
Ship traffic density by ship types >300 GT in 2024
Data derived from satellited-based AIS data (S-AIS) from exactEarth/Spire
0 750 1,500 3,000
Kilometres


Cargo >300 GT
Container >300 GT
Dry bulk >300 GT
Ferry >300 GT
Fishing >300 GT
Government/research >300 GT
Others/special ships >300 GT
Passenger (cruise) >300 GT
Pleasure vessels >300 GT
Tanker >300 GT
Tugs (barge) >300 GT
Difference in yearly trips per 50 km²
<= -500
-499 to -250
-249 to -100
-99 to -50
-49 to -10
+10 to +49
+50 to +99
+100 to +249
+250 to + 499
>= +500
Difference in ship traffic density by ship types >300 GT 2023 vs 2013
Blue = more in 2013
Red = more in 2023
Data derived from satellited-based AIS data (S-AIS) from exactEarth/Spire
0 750 1,500 3,000
Kilometres

Spatial Shipping Hot Spot Risk Areas

Total unique vessels = increasing
Total kilometers travelled = increasing
Accident rate = decreasing (varies regionally)





Relative difference from North of 60 2012 to 2019 accident rate for all non-commercial vessels >300 GT

<=-80% | -79.9 to -60% | -59.9 to -40% | -39.9 to 20% | -19.9 to 0% | No data/accidents
+0.01 to +20% | +20.1 to +40% | +40.1 to +60% | +60.1 to +80% | >= +80.1% | North of the 60th (N60) parallel

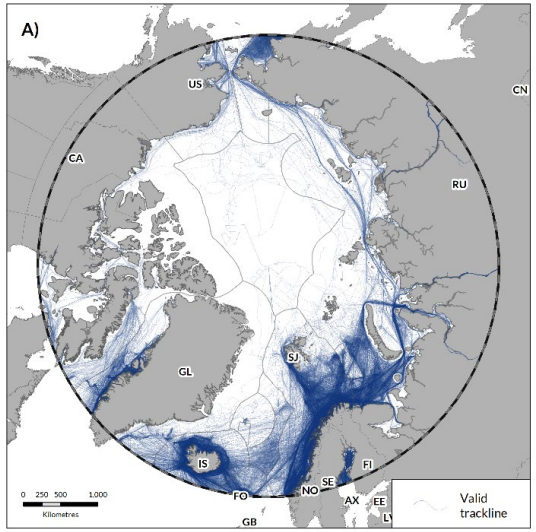
0 362.5725 1,450
Kilometres

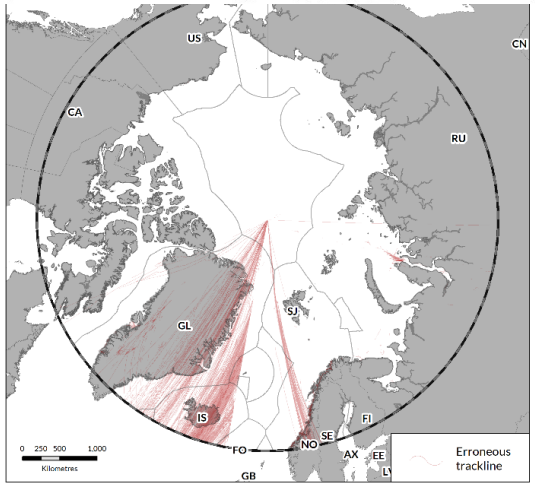
High Risk Clusters

Arctic Shipping Trends Data Challenges

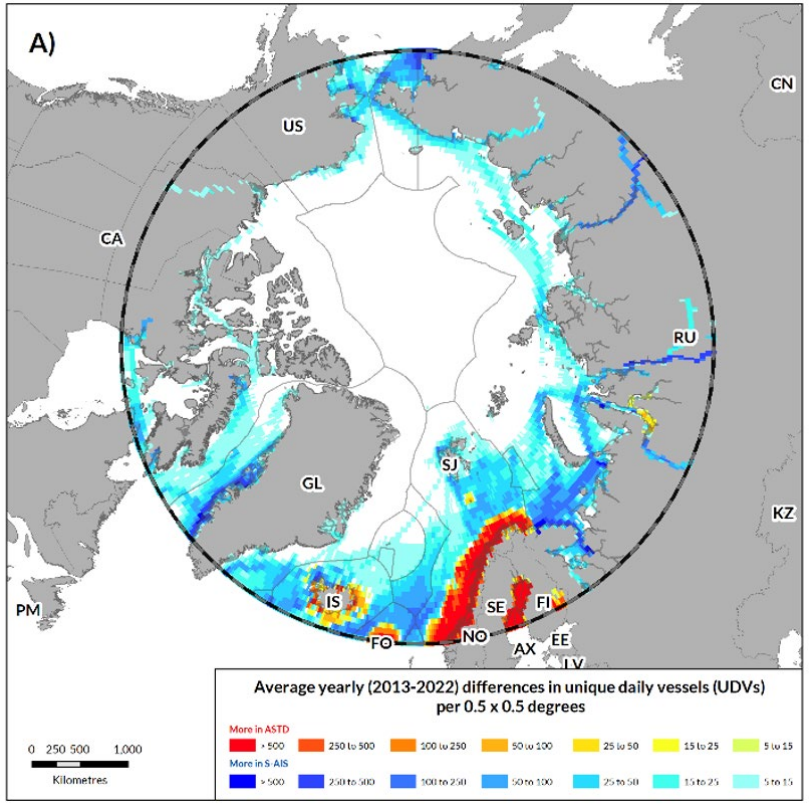
A) cleaned S-AIS tracklines in N62 for the year 2020 as an example.

B) B) S-AIS tracklines identified as spoofing vessels



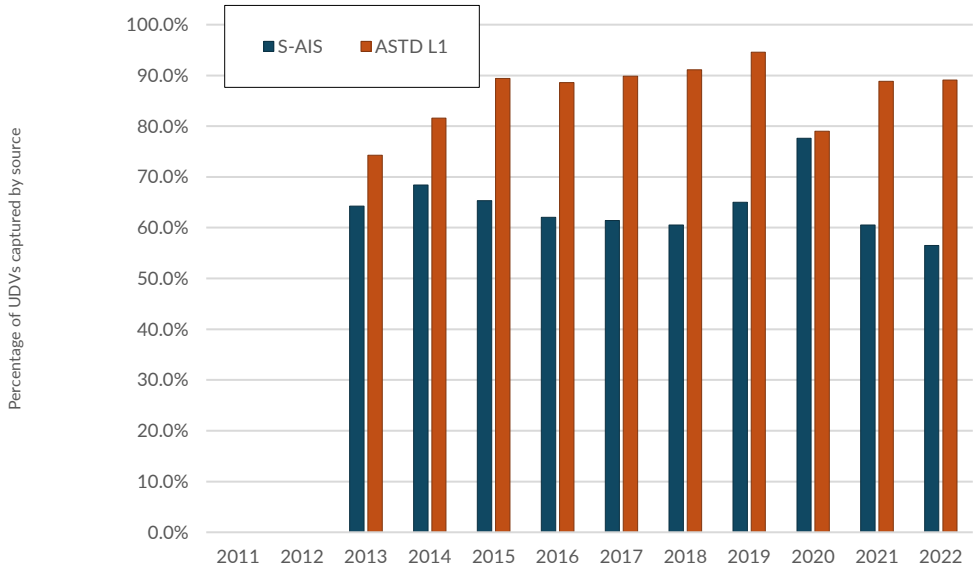


Yearly average (2013-2022) difference in unique daily vessels between ASTD L1 and S-AIS datasets per 0.5 x 0.5-degrees; grid cells in red represent more UDV detected in the ASTD L1 data, while grid cells in blue represent more UDV detected in S-AIS

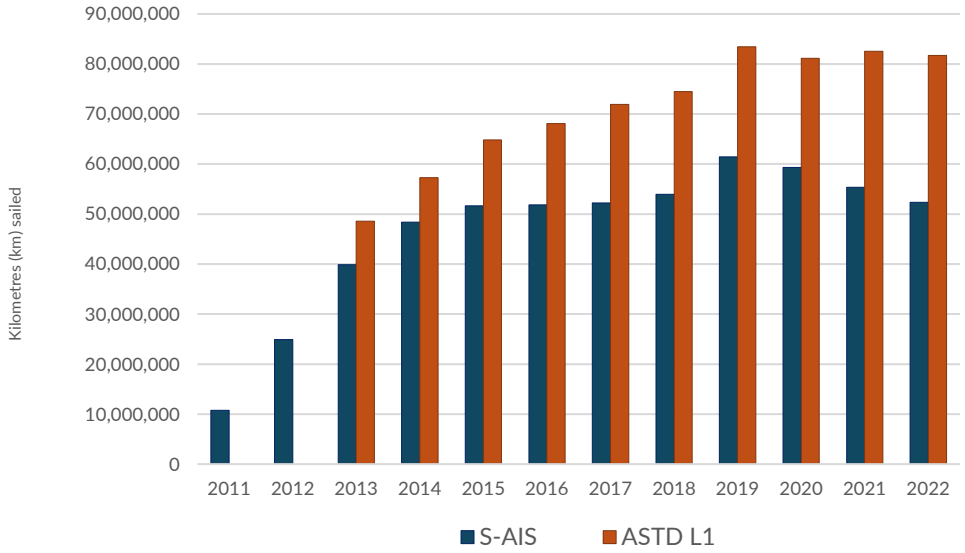


Nicol et al. 2025

Percentage (%) of unique daily vessels captured by S-AIS and ASTD L1 North of the 62nd parallel from 2011 to 2022.



Derived kilometres (km) sailed by S-AIS and ASTD L1 North of the 62nd parallel from 2011 to 2022.



Thank you!

Dr Jackie Dawson
jackie.dawson@uottawa.ca
www.espg.ca

ESPG Environment, Society and Policy Group

uOttawa